ENGINE—APPROVED HEAD GASKET REPLACEMENT SERVICE PROCEDURES	Article No. 02-1-4

FORD: 1992-1997 THUNDERBIRD
1992-2002 CROWN VICTORIA, ESCORT, MUSTANG, TAURUS
1995-2000 CONTOUR
2000-2002 FOCUS
1995-2002 WINDSTAR
1997-2002 E SERIES, EXPEDITION, EXPLORER, F-150
1999-2002 SUPER DUTY F SERIES
2000-2002 EXCURSION
2001-2002 ESCAPE

LINCOLN: 1992-2002 CONTINENTAL, TOWN CAR
2000-2002 LS
1998-2002 NAVIGATOR
2002 BLACKWOOD

MERCURY: 1992-1997 COUGAR
1992-1999 TRACER
1992-2002 GRAND MARQUIS, SABLE
1995-2000 MYSTIQUE
1999-2002 COUGAR
1993-2002 VILLAGER
1997-2002 MOUNTAINEER

ISSUE

Some engine assemblies with aluminum cylinder heads repaired in-vehicle have been found to leak coolant and/or oil from the cylinder head gasket area due to particle contamination or scratches and dents at the head to block mating surfaces.

ACTION

Refer to the following Service Information for recommended gasket sealing surface preparation procedures. If the engine diagnosis points to a faulty head gasket (leaks oil, leaks coolant, oil or coolant contamination, or air in coolant), thorough cleaning procedures are necessary for a successful head gasket repair. Diagnosis and post-repair confirmation can be done successfully utilizing leak detection methods such as a black light dye process.

SERVICE PROCEDURE

Aluminum Cylinder Head - Removal From Engine

WARNING

ALUMINUM CYLINDER HEADS ARE HIGHLY POLISHED AND MUST BE HANDLED WITH EXTREME CARE. AFTER REMOVING THE HEAD BOLTS, WITHDRAW THE HEAD FROM THE ENGINE COMPARTMENT. USE EXTREME CARE NOTING THAT THE GASKET SURFACE OF THE HEAD COULD BE DAMAGED ON THE WAY OUT OF THE VEHICLE IF ALLOWED TO HIT AGAINST OTHER ENGINE COMPONENTS. ONCE REMOVED, PLACE THE HEAD ON A BENCH, GASKET SIDE UP, USING A PIECE OF CLEAN CARDBOARD ON THE BENCH SURFACE UNDERNEATH. DO NOT SLIDE THE HEAD GASKET SURFACE ALONG THE BENCHTOP OR DAMAGE WILL OCCUR.

NOTE

SOME ENGINES REQUIRE A "DE-TORQUING" PROCEDURE OF THE CYLINDER HEAD BOLTS.

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CONSULT WITH THE APPROPRIATE MODEL YEAR WORKSHOP MANUAL FIRST, AS THIS PROCEDURE IS USUALLY DONE ON FOUR-VALVE ENGINES. HOWEVER, ALL ALUMINUM HEADS ARE PRONE TO WARPAGE DURING REMOVAL IF TAKEN OFF WHEN HOT OR WARM. THE ENGINE SHOULD BE COOLED SUFFICIENTLY BY THE TIME THE HEADS ARE ACCESSIBLE DURING TEARDOWN. ALLOW AN ADEQUATE COOL DOWN PERIOD.

NOTE
IN SOME ENGINE APPLICATIONS, IT MAY BE NECESSARY TO "RUBBER BAND" A FEW OF THE HEAD BOLTS TOGETHER DUE TO TIGHT CLEARANCES OF THE HEAD NEAR HEATER/AC PLENUMS AND THE BODY COWL. THIS WILL ALLOW THE BOLTS TO CLEAR THE BLOCK DECK AS THE HEAD IS WITHDRAWN FROM THE VEHICLE.

Place **CLEAN** shop towels over the cylinder block bores to prevent further contamination from reaching internal parts and surfaces. If a towel is to be removed, do it **SLOWLY** . Debris collected on the towel can fall on the engine if care is not exercised as it is removed.

Gasket surfaces are machined to near-mirror surface quality. Scratches, dents, gouges, and other impressions may allow oil, coolant, and compression gases to leak into other sealed cavities and/or the atmosphere. Surface imperfections deteriorate the head gasket's ability to seal effectively. Contamination on these surfaces will cause leaks from "shimming" the head even under full head bolt torque. Contamination will also cause marks on the polished gasket surface, creating a leak path.

Inspection For Damage And Cleaning Procedure

The overall straightness of the cylinder head and block face is held within very tight specification. Defects known as "waviness" or "depressions" cannot be surface machined out with ordinary equipment. Machining must meet manufacturer's specification on surface finish quality and be less than 0.001″ inch (0.025 mm) level as measured under a **known quality straightedge** using a feeler gauge. If the engine has been overheated the cylinder head may have been damaged or warped. Re-surfacing will not correct this damage.

The head must not contain any impressions on its sealing surface deeper than 0.001″ (0.025 mm). There must not be any scratches or gouges present, especially those which track to another sealing cavity or to the atmosphere.

Ensure that the mating cylinder block surface is completely free of solid contamination, corrosion, and fluids. Use Motorcraft Metal Surface Cleaner, F4AZ-19A536-RA or equivalent, to rid the surface of any material, which could later interfere with the gasket's sealing ability.

WARNING
UNDER NO CIRCUMSTANCES ARE ANY ALUMINUM GASKET SURFACES TO BE CLEANED USING RAZOR BLADES, ROTARY ABRASIVE DEVICES INCLUDING ROLOC AND 3M BRANDED SCOTCHBRITE PRODUCTS, ROTARY WIRE BRUSHES, SINGLE HANDLED WIRE BRUSHES, HAND ABRASIVE SUCH AS SANDPAPER OR EMERY CLOTH, OR ANY CARBON STEEL BLADE. THESE TOOLS ARE PROVEN TO CUT AND DAMAGE ALUMINUM AND WILL DISRUPT THE POLISHED FINISH. ABRASIVE PARTICLES ARE ALSO SUSCEPTIBLE TO ENTERING THE ENGINE CAVITIES AND MAY CAUSE INTERNAL ENGINE DAMAGE. THE ONLY TOOLS ACCEPTABLE ARE PLASTIC AND WOOD SCRAPERS COMBINED WITH USE OF MOTORCRAFT METAL SURFACE CLEANER, F4AZ-19A536-RA, OR EQUIVALENT SOLVENT.

The original head bolts are to be discarded. Only new head bolts are acceptable for use in a head gasket repair. Use of old bolts can cause clamp load failure because they were stretched from the last torquing procedure either from the plant or a previous field repair. The new head bolt shanks and threads should be thoroughly cleaned, dried, and lightly lubricated. Oil the bolts and let oil drip for a minimum of 5 minutes. Use of too much oil may cause hydraulic lockup in the bolt hole.

The corresponding bolt holes in the cylinder block must be free of contamination consisting of dirt particles, coolant, and oil. Bolts can mechanically or hydraulically lock on top of these materials and cause for poor clamping of the cylinder head gasket. Use very lightly compressed air to blow out the bolt holes. Use care to prevent debris from scattering over the internal engine surfaces.

To clean the cylinder block gasket surface, a plastic or wood scraper in combination with Motorcraft Metal Surface Cleaner, F4AZ-19A536-RA, or equivalent solvent can be used with a portable shop vacuum. Move the scraper toward the vacuum nozzle to direct loosened material quickly away from the block surface.

It is not necessary to expect aluminum head surfaces to be shiny and bright after the vehicle has been in service. Sanding, scraping, or polishing will create surface depressions that cause leaks. The surface is expected to be flat within 0.001 ″ (0.025 mm) and free of dirt, metal chips, and liquid contaminants. Any staining of the metal surface is considered normal.

Machining of aluminum heads and blocks is an unacceptable field practice for quality control purposes of flatness and surface finish standards. Removing material will also raise the risk of internal reciprocating components striking each other, as well as changing the emissions calibration of the engine. Machining practices of aluminum blocks and heads are not reimbursable. Ford Remanufacturers are authorized to perform this procedure and have the necessary equipment to put the specified surface finish on the head within original factory limits.

Installation Onto The Cylinder Block Surface

WARNING
ANY GASKET SEALING AIDS WHICH ARE NOT LIMITED TO, AND INCLUDE AVIATION CEMENT, COPPER SPRAYS, OR GLUES ARE NOT TO BE USED. THE GASKET MUST BE FITTED DRY WITHOUT SUPPLEMENTAL SEALING AIDS. ATTEMPTS TO SEAL WITH THESE COMPOUNDS WILL CAUSE HEAD SHIMMING AND/OR NEW LEAK PATHS. THESE COMPOUNDS HAVE ALSO BEEN KNOWN TO CAUSE EMISSIONS SENSORS TO FAIL THEREBY TRIPPING FAULT CODES IN THE PCM.

Ensure that all surfaces of the cylinder block, cylinder head, and gasket are fully free of any solid or liquid contamination and have been prepared for assembly using Motorcraft Metal Surface Cleaner, F4AZ-19A536-RA, or equivalent solvent. Also ensure that any residual dirt on the engine was removed successfully from the surrounding areas of the block to head face of the head and the block.

Using Workshop Manual procedures, assemble the head to the block carefully. Do not allow the gasket surfaces to scrape or bang against other engine parts. The head may pick up material along the way, if this is allowed. Once in place, with dowels properly aligned to the head and ensuring that the head is fully square, seated, and resting on the block deck in a level condition, install new "torque-to-yield" cylinder head bolts to a finger tight condition. Refer to the appropriate model year Workshop Manual for the procedure to torque the head bolts in the correct sequence and at the intervals specified. Failure to follow these instructions may result in a warped head, and damaged head bolt threads.

Do not use "cheater pipes" on torque wrenches. Use a known quality, calibrated torque wrench. Pull evenly and steady on each bolt. Do not exert side pressures on the bolt hex, which can disrupt the integrity of the bolt head and give a false reading on the final torque value of the fastener.

Once the engine is fully reassembled, be sure that fluid levels have been topped off as required for the operation. When the engine is ready for starting, ensure that the engine oil pump is (was) primed properly and is in a "ready to start" condition with all engine oil galleys having been charged with oil.

WARNING
DO NOT RACE THE ENGINE UPON INITIAL STARTUP. MAINTAIN ENGINE IDLE TO BUILD OIL PRESSURE. USE OF A MECHANICAL PRESSURE GAUGE WILL ENSURE THAT OIL FLOW HAS DEVELOPED ON THE PUMP'S PRESSURE SIDE. A DEFECTIVE GAUGE OR LIGHT IN THE INSTRUMENT PANEL WILL NOT VERIFY PRESSURE HAS BEEN OBTAINED. SEVERE INTERNAL ENGINE DAMAGE WILL RESULT IF PRESSURE IS NOT OBSERVED AND STEADILY MAINTAINED.

Allow a warm up period, which brings the engine to normal operating temperature. Check for leaks around the gasket joints of the head, front cover, and associated component parts. The vehicle should be road tested within normal guidelines to raise the engine to operating temperature and verify the repair is sound.

Shut down the engine, allow for a cool-down and drain-back period to enable fluids to return to levels. Check fluid levels using owner guide recommendations (dipstick, sight gages, etc.).

To sum up, a successful leak repair depends on:

- Accurate diagnosis of the leak using Ford-recommended test equipment and procedures
- Surface preparation and gasket installation using the approved tools, cleaners, and methods
- Exercising CARE and CLEANLINESS during disassembly/assembly of components
- Use of genuine Ford OEM replacement parts
- Use of a Ford Remanufacturer

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 401000, 402000, 403000, 499000